

Ahmed Aziz Ben Haj Hmida

Machine Learning Engineer (EPFL, MSc Data Science, GPA 5.3/6) specialising in computer vision, LLMs, and agentic AI systems. Track record of building production-ready ML pipelines with significant, measurable performance gains. Hands-on approach across the full stack from data to deployment.

azizbenhajhmida@gmail.com | +41 78 812 04 05 | Lausanne, Switzerland
ahmed-aziz-ben-haj-hmida | azizbenhaj | azizbenhaj.github.io

Professional Experience

New Wind Ventures SA — Geneva, Switzerland
AI/ML Engineer — Oct 2026 – Present

- Leading data migration and integration of source systems into the production platform.
- Deploying AI models for automated mapping, quality checks, and workflow optimisation.

New Wind Ventures SA — Geneva, Switzerland
Data Engineer Internship — May 2026 – Sep 2026

- Built SQL and Python pipelines to ingest, clean, and structure heterogeneous datasets.
- Implemented data-quality checks that standardised sources for production integration.

Rollomatic SA — Le Landeron, Switzerland
Master's Thesis – Curve-Based CNC Tool Similarity Retrieval – ***Grade 6/6*** — Aug 2025 – apr 2026

- Built an end-to-end CNC tool retrieval system combining geometric curves and visual embeddings (**+11%** accuracy).
- Designed indexing and benchmarking pipelines for integration into industrial R&D workflows.

Supervisors: Prof. Jean-Philippe Thiran – jean-philippe.thiran@epfl.ch, Emna Maghrebi – e.maghrebi@rollomatic.ch

Rollomatic SA — Le Landeron, Switzerland
Machine Learning Internship — Feb 2025 – Aug 2025

- Curated a 5000+ CNC tool dataset and built hybrid retrieval pipelines, cutting manual search time by **96%**.

Supervisor: Emna Maghrebi – e.maghrebi@rollomatic.ch

Technical Skills

Core Expertise: Python, PyTorch, scikit-learn, NumPy, Pandas
LLMs & NLP: HuggingFace Transformers, PEFT, LoRA, RLHF, Quantization
Agentic AI Systems: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, MCP, RAG systems
Computer Vision: OpenCV, torchvision, Vision Transformers, CNNs, ResNet, scikit-image
Data Engineering: PySpark, PostgreSQL, SQL, FAISS, Milvus, ChromaDB
MLOps & DevOps: Docker, Kubernetes, MLflow, W&B, DVC, FastAPI, CI/CD, AWS, GCP
Visualization: Matplotlib, Seaborn, Plotly, Power BI
Other Languages: Java, SQL, Scala

Education

EPFL – École Polytechnique Fédérale de Lausanne — Lausanne, Switzerland
M.Sc. in Data Science — Sep 2023 – Mar 2026
GPA: 5.3/6
Relevant Coursework: Machine Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Analysis, Database Systems, Data Science for Business

Technical University of Denmark (DTU) — Lyngby, Denmark
3rd Year of Bachelor - Exchange — Aug 2022 – Jul 2023
GPA: 5.35/6

EPFL – École Polytechnique Fédérale de Lausanne — Lausanne, Switzerland
B.Sc. in Communication Systems — Sep 2020 – Jul 2023
GPA: 5.1/6

Pioneer School of Nabeul — Nabeul, Tunisia
Baccalaureate in Mathematics — Sep 2018 – Jun 2019

Research Projects

Unsupervised Reverse Knowledge Distillation and Model Merging — Aug 2024 – Feb 2025
EPFL LTS4 Lab

- Designed a reverse distillation framework transferring knowledge from fine-tuned specialist models into larger architectures, preserving general capabilities while boosting niche-task performance.
- Applied model merging techniques (weight interpolation, task vectors) to combine specialised and general models

without full retraining.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Discovering promising new startups through evaluating teams' potentials Sep 2023 – Dec 2023
EPFL ENTC Lab

- Applied NLP and multimodal ML to infer MBTI personality types and emotional signals from pitch recordings, creating behavioural features from unstructured audio/text data.
- Built ensemble classification models predicting startup success likelihood from team dynamics signals, achieving actionable rankings across a labelled cohort of entrepreneurial teams.

Supervisor: Prof. Davide Bavato

Personal & Academic Projects

AI-Powered Food & Nutrition Assistant May 2026 - Aug 2026

- Built a React frontend and Python backend with a LangChain ReAct agent for natural language meal logging, GPT-4o vision for photo-based meal analysis, and ChromaDB semantic search over 200k+ recipes enriched with USDA nutritional data.
- Tracked daily macro intake with personalised targets (Mifflin–St Jeor), automated email digests via an MCP tool, and deployed to AWS with a GitHub Actions CI/CD pipeline and CloudWatch monitoring.

LLM Agent–Driven Car Search Assistant Mar 2026 - May 2026

- Architected a full-stack conversational search tool using LangGraph + LangChain, enabling users to describe a vehicle in natural language and automatically populate structured queries against a PostgreSQL database.
- Generated ranked listings with match scores, interactive Plotly/Matplotlib charts, and auto-exported PDF comparison reports for saved car profiles.

LLM Teaching Assistant for MCQ Answering Feb 2024 - Jun 2024

- Fine-tuned Qwen1.5 using SFT and DPO with 4-bit quantization and RAG retrieval, balancing response quality and inference efficiency for an educational setting.
- Built an evaluation pipeline using LLM-as-a-Judge and cross-validation to measure pedagogical accuracy, enabling systematic model comparison and iteration.

Coin Detector and Counter Feb 2024 - Jun 2024

- Built a multi-stage vision pipeline combining CNN-based segmentation and ResNet50 / ViT classification to detect and identify coins across varied and uncontrolled backgrounds.

RL for Motor Skill Transfer Feb 2024 - Jun 2024
EPFL Alexander Mathis Lab

- Trained an expert RL agent on the Myosuite baoding balls task and distilled its motor control strategies to novice agents via behaviour cloning, improving sample efficiency and final performance on a complex continuous-control benchmark.

Machine Learning Operations Jan 2023 - Mar 2023

- Built a production-grade ML pipeline integrating PyTorch, Docker, Kubernetes, MLflow, DVC, AWS, and GCP — covering model versioning, containerised training, and automated deployment.

Certificates

Data Analysis Certificate – Mayerfeld Consulting Jun 2026 – Aug 2026
IBM RAG and Agentic AI – IBM Apr 2026 - Jun 2026
DevOps and AI on AWS – AWS May 2026 - Jun 2026
Structuring Machine Learning Projects – DeepLearning.AI Sep 2023 - Oct 2023
Certificate in Artificial Intelligence – GOMYCODE Jun 2023 – Sep 2023

Leadership and Volunteering

EPFL Coaching – CMS section Coach Sep 2023 - Jun 2024
Rotary International – Member Jul 2018 - Apr 2019
Tunisian Red Crescent – Volunteer May 2018 - Apr 2019

Languages

Arabic (Native), French (Native/C2), English (C1), German (A2)